

Ankit

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Summary

M.Sc. candidate in Computational Mathematics and Data Science at NIT Rourkela with working knowledge in Python, C++, statistics, machine learning basics and deep learning basics. Academic background covers linear algebra, probability, numerical analysis, and optimization.

Education

National Institute of Technology, Rourkela Aug 2025 – Present
M.Sc. in Computational Mathematics and Data Science (Certificate) CGPA: 7.95

Sh. Rameshwar Lal Sharma Teacher Training College, Pilani 2021 – 2025
B.Sc. – B.Ed (Certificate) CGPA: 7.23

Research Experience

Numerical Analysis Laboratory Aug 2025 – Nov 2025
Python, NumPy, SciPy

- Implemented algorithms for nonlinear equations, linear systems, and ODE models
- Performed convergence and error analysis for numerical stability
- Benchmarked efficiency of iterative computational methods

Statistics Laboratory Jan 2026 – Present
C++

- Built statistical computation programs from scratch (no STL)
- Implemented simulation models for probability distributions
- Computed descriptive and inferential statistics efficiently.

Projects

Sleep Breathing Analysis
Python, TensorFlow, Keras, NumPy, SciPy, Matplotlib

- Built an end-to-end pipeline to classify sleep breathing events (Hypopnea, Obstructive Apnea) from PSG signals across 5 participants.
- Applied 4th-order Butterworth bandpass filter (0.17–0.4 Hz), segmented into 30-second windows with 50% overlap, and trained a 1D CNN evaluated via Leave-One-Participant-Out Cross-Validation with classification reports and confusion matrices.
- Generated per-participant PDF diagnostic reports with annotated PSG signal overlays.

Heatwave Prediction in Rajasthan
Python, Pandas, NumPy, Scikit-learn, Matplotlib

- Performed exploratory data analysis (EDA) on multi-year meteorological datasets
- Engineered heat index features capturing temperature–humidity interactions
- Implemented and evaluated ML models including Random Forest and Support Vector Machine (SVM) for heatwave prediction

Technical Skills

Programming: Python | **APIs:** FastAPI
Data Analysis: Pandas, NumPy, Matplotlib-seaborn visualization, EDA, Statistical Modeling
Machine Learning: Regression, Decision Trees, ensemble, Model Evaluation
Deep Learning: TensorFlow, Keras, 1D CNN
Database: MySQL **Tools:** Git, GitHub

Certification

Python Essentials – Cisco (2025) (Certificate)